

CS3103-1: THEORY OF COMPUTATION

Multiple Choice Questions — Compiled with Answers (Side-by-Side)

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UNIT - I: Multiple Choice Questions

1. Which is correct?

- a) $a^+ = a^* \cdot a^*$
- b) $a^* = a^+ \cdot a^+$
- c) $a^+ = a^* \cdot a$ ✓
- d) $a^* = a^+ \cdot a^*$

CORRECT ANSWER

Answer: C

$a^+ = a^* \cdot a$

2. Which of the following is not a part of 5-tuple finite automata?

- a) **Input alphabet** ✓
- b) Transition function
- c) Initial State
- d) Output Alphabet

CORRECT ANSWER

Answer: A

Input alphabet

3. $(a, b)^*$ means

- a) Any combination of a, b including null
- b) **Any combination of a, b excluding null** ✓
- c) Any combination of a, b, but 'a' will come first
- d) None of these

CORRECT ANSWER

Answer: B

Any combination of a, b excluding null

4. $(a, b)^+$ means

- a) Any combination of a, b including null
- b) **Any combination of a, b excluding null** ✓
- c) Any combination of a, b, but 'a' will come first
- d) None of these

CORRECT ANSWER

Answer: B

Any combination of a, b excluding null

5. What is the language generated by the grammar $S \rightarrow aSb$, $S \rightarrow A$, $A \rightarrow aA$

- a) $a^m b^m$
- b) $\emptyset$ ✓
- c) $a^n b^m$
- d) $a^m b^m$

CORRECT ANSWER

Answer: B

$\emptyset$

6. The language generated by the grammar $S \rightarrow 0S0$, $S \rightarrow 1S1$, $S \rightarrow 0$, $S \rightarrow 1$ is

- a) Even palindrome of 0 and 1
- b) Odd palindrome of 0 and 1
- c) Any combination of 0 and 1
- d) **None of these** ✓

CORRECT ANSWER

Answer: D

None of these

7. Which of the following grammar generates strings with any number of 1's?

- a) $S \rightarrow 1A$, $A \rightarrow \epsilon$
- b) **$S \rightarrow 1S$, $S \rightarrow \epsilon$** ✓
- c) $S \rightarrow S1$, $S \rightarrow \epsilon$
- d) (b) & (c)

CORRECT ANSWER

Answer: B

$S \rightarrow 1S$, $S \rightarrow \epsilon$

8. Which of the following is true?

- a) $(ab)^2 = a^2b^2$
- b) $(ab)^2 = abab$
- c) **$(ab)^2 = aabb$** ✓
- d) None of these

CORRECT ANSWER

Answer: C

$(ab)^2 = aabb$

9. If $\Sigma = \{1\}$, then $\Sigma^* - \Sigma^+$ is

- a) 1^+
- b) **$\{1\}$** ✓
- c) $\{\epsilon\}$
- d) $\{\epsilon, 1, 11, \dots\}$

CORRECT ANSWER

Answer: B

$\{1\}$

10. A language set contains

- a) Alphabet
- b) String
- c) Set
- d) **None of these** ✓

CORRECT ANSWER

Answer: D

None of these

11. In automata, why it is called formal language?
a) **Some alphabet forms a string** ✓
b) Some strings form a language
c) Well-defined use of symbols is significant
d) Only the form of the string generated by alphabets is significant.

CORRECT ANSWER
Answer: A
Some alphabet forms a string

12. The language constructed from the grammar $S \rightarrow aSbb/aAbb, A \rightarrow a$ is
a) $a^{n+1} b^{2n}$
b) $a^n b^{2n}$
c) $a^n b^{2n-1}$
d) **$a^{n+1} b^{2n+1}$** ✓

CORRECT ANSWER
Answer: D
 $a^{n+1} b^{2n+1}$

13. The language generated by the grammar $S \rightarrow aSa/aBa, A \rightarrow Ba/b$ is
a) $a^n b a^n$
b) $a^n b a^{n+1}$
c) a^{2n}
d) Φ

CORRECT ANSWER
Answer not provided in source

UNIT-1: Finite Automata — Multiple Choice Questions

1. A language L from a grammar $G = \{V_N, \Sigma, P, S\}$ is
a) Set of symbols over V_N
b) **Set of symbols over Σ** ✓
c) Set of symbols over P
d) Set of symbols over S

CORRECT ANSWER
Answer: B
Set of symbols over Σ

2. Which is true for $\delta(q, ab)$
a) $\delta(q, a) \cup \delta(q, b)$
b) **$\delta(\delta(q, a), b)$** ✓
c) $\delta(q, a), b$
d) $\delta(q, a) \cap \delta(q, b)$

CORRECT ANSWER
Answer: B
 $\delta(\delta(q, a), b)$

3. The transitional function of a DFA is
a) **$Q \times \Sigma \rightarrow Q$** ✓
b) $Q \times \Sigma \rightarrow 2^Q$
c) $Q \times \Sigma \rightarrow 2^n$
d) $Q \times \Sigma \rightarrow Q^n$

CORRECT ANSWER
Answer: A
 $Q \times \Sigma \rightarrow Q$

4. The transitional function of an NFA is
a) $Q \times \Sigma \rightarrow Q$
b) **$Q \times \Sigma \rightarrow 2^Q$** ✓
c) $Q \times \Sigma \rightarrow 2^n$
d) $Q \times \Sigma \rightarrow Q^n$

CORRECT ANSWER
Answer: B
 $Q \times \Sigma \rightarrow 2^Q$

5. The maximum number of states of a DFA converted from an NFA with n states is
a) n
b) n^2
c) **2^n** ✓
d) None of these

CORRECT ANSWER
Answer: C
 2^n

6. A string after full traversal is called not accepted by an NFA if it results in
a) Some non-final states
b) **All non-final states** ✓
c) A single non-final state
d) Some final states

CORRECT ANSWER
Answer: B
All non-final states

7. An NFA with a set of states Q is converted to an equivalent DFA with a set of states Q'. Find which is true.
a) $Q' = Q$
b) $Q' \subseteq Q$
c) $Q \subseteq Q'$
d) **None of these** ✓

CORRECT ANSWER
Answer: D
None of these

8. The basic limitations of a finite state machine is
a) **It cannot remember arbitrarily large amount of information** ✓
b) It cannot remember state transitions
c) It cannot remember grammar for a language
d) It cannot remember language generated from a grammar

CORRECT ANSWER
Answer: A
It cannot remember arbitrarily large amount of information

9. The string WW^R is not recognized by any FSM because
- a) An FSM cannot remember arbitrarily large amount of information
 - b) **An FSM cannot fix the mid-point ✓**
 - c) An FSM cannot match W with W^R
 - d) An FSM cannot remember the first and last inputs.

CORRECT ANSWER

Answer: B

An FSM cannot fix the mid-point

10. A finite automata recognizes
- a) Any language
 - b) Context sensitive language
 - c) Context-free language
 - d) **Regular language ✓**

CORRECT ANSWER

Answer: D

Regular language

11. Which is true for a dead state?
- a) It cannot be reached anytime
 - b) There is no necessity of the state
 - c) **If control enters, there is no way to come out from the state ✓**
 - d) If control enters, FA is dead

CORRECT ANSWER

Answer: C

If control enters, there is no way to come out from the state

12. The language accepted by the given FA is (diagram: states A, B, C with a/b transitions)
- a) $(ab)^*$
 - b) bb^*a
 - c) $b(ba)^*a$
 - d) **Null ✓**

CORRECT ANSWER

Answer: D

Null

13. In the previous FA, B is called
- a) Dead state
 - b) **Inaccessible state ✓**
 - c) Both a and b
 - d) None of these

CORRECT ANSWER

Answer: B

Inaccessible state

14. Which is true for the inaccessible state?
- a) **It cannot be reached anytime ✓**
 - b) There is no necessity of the state
 - c) If control enters, there is no way to come out from the state
 - d) If control enters, FA is dead

CORRECT ANSWER

Answer: A

It cannot be reached anytime

15. Which type of string is accepted by the following finite automata? (single circle, start = final, no transitions shown)
- a) All string
 - b) **Null string ✓**
 - c) No string
 - d) All of the above

CORRECT ANSWER

Answer: B

Null string

GATE Questions — Finite Automata

1. Consider the strings $u = abbaba$, $v = bab$, and $w = aabb$. Which of the following statement is true for the given transitional system?
- a) The automaton accepts u and v but not w .
 - b) The automaton accepts each of u , v , and w .
 - c) The automaton rejects each of u , v , and w .
 - d) **The automaton accepts u but rejects v and w . ✓**

CORRECT ANSWER

Answer: D

The automaton accepts u but rejects v and w .

2. Consider the transitional system. Which of the states are to be marked as starting state and final state, respectively, so as to turn the above system into a DFA that accepts all strings having odd number of zeros and even number of 1's?
- a) q_0, q_2
 - b) q_0, q_1
 - c) **q_1, q_2 ✓**
 - d) None of these

CORRECT ANSWER

Answer: C

q_1, q_2

3. Consider the following DFA in which S_0 is the start state and S_1 and S_3 are the final states. Which one is true?
- a) All strings of x and y .
 - b) All strings of x and y which have either an even number of x and even number of y or an odd number of x and odd number of y .
 - c) All strings of x and y which have an equal number of x and y .
 - d) **All strings of x and y which have either an even number of x and odd number of y or an odd number of x and even number of y . ✓**

CORRECT ANSWER

Answer: D

All strings of x and y which have either an even number of x and odd number of y or an odd number of x and even number of y .

4. Let N be an NFA with n states and let M be the minimized DFA with m states recognizing the same language. Which of the following is NECESSARILY true?

- a) **$m \leq 2^n$** ✓
- b) $n \leq m$
- c) M has one accept state
- d) $m = 2^n$

CORRECT ANSWER

Answer: A

$m \leq 2^n$

5. If the final state and non-final states in the following DFA are interchanged, then which of the following languages over the alphabet (a, b) will be accepted by the new DFA?

- a) **Set of all strings that do not end with ab** ✓
- b) Set of all strings that begin with either an a or a b.
- c) Set of all strings that do not contain the substring ab.
- d) The set described by the regular expression $b^*aa^*(ba)^*b^*$

CORRECT ANSWER

Answer: A

Set of all strings that do not end with ab

6. A finite state machine with the following state table has a single input x and a single output z. If the initial state is unknown, then the shortest input sequence to reach the final state C is:

- a) 01
- b) **10** ✓
- c) 101
- d) 110

CORRECT ANSWER

Answer: B

10

7. Which of the following sets can be recognized by a deterministic finite state automaton?

- a) **The numbers 1, 2, 4, 8, . . . 2^n written in binary.** ✓
- b) The numbers 1, 2, 4, . . . 2^n written in unary.
- c) The set of binary string in which the number of zeros is the same as the number of ones.
- d) The set {1, 101, 11011, 1110111, . . . }

CORRECT ANSWER

Answer: A

The numbers 1, 2, 4, 8, . . . 2^n written in binary.

8. Consider the regular expression $(0 + 1)(0 + 1) \dots N$ times. The minimum state FA that recognizes the language represented by this regular expression contains

- a) n states
- b) **$(n + 1)$ states** ✓
- c) $(n + 2)$ states
- d) None of the above

CORRECT ANSWER

Answer: B

$(n + 1)$ states

9. What can be said about a regular language L over {a} whose minimal finite state automation has two states?

- a) **L must be $\{a^n \mid n \text{ is odd}\}$** ✓
- b) L must be $\{a^n \mid n \text{ is even}\}$
- c) L must be $\{a^n \mid n \geq 0\}$
- d) Either L must be $\{a^n \mid n \text{ is odd}\}$ or L must be $\{a^n \mid n \text{ is even}\}$

CORRECT ANSWER

Answer: A

L must be $\{a^n \mid n \text{ is odd}\}$

10. The smallest FA which accepts the language $\{x \mid \text{length of } x \text{ is divisible by } 3\}$ has

- a) 2 states
- b) **3 states** ✓
- c) 4 states
- d) 5 states

CORRECT ANSWER

Answer: B

3 states

11. Consider the following deterministic finite state automaton M. Let S denote the set of seven-bit binary strings in which the first, fourth, and last bits are 1. The number of strings in S that are accepted by M is

- a) **1** ✓
- b) 5
- c) 7
- d) 8

CORRECT ANSWER

Answer: A

1

12. The following finite state machine accepts all those binary strings in which the number of 1's and 0's are, respectively,

- a) **divisible by 3 and 2** ✓
- b) odd and even
- c) even and odd
- d) divisible by 2 and 3

CORRECT ANSWER

Answer: A

divisible by 3 and 2

13. Consider the machine M. The language recognized by M is: a) $\{w \in \{a, b\}^* \mid \text{every } a \text{ in } w \text{ is followed by exactly two } b\text{'s}\}$ ✓ b) $\{w \in \{a, b\}^* \mid \text{every } a \text{ in } w \text{ is followed by at least two } b\text{'s}\}$ c) $\{w \in \{a, b\}^* \mid w \text{ contains the substring 'abb'}\}$ d) $\{w \in \{a, b\}^* \mid w \text{ does not contain 'aa' as a substring}\}$	CORRECT ANSWER Answer: A $\{w \in \{a, b\}^* \mid \text{every } a \text{ in } w \text{ is followed by exactly two } b\text{'s}\}$
14. A minimum state deterministic FA accepting the language $L = \{w \mid w \in \{0, 1\}^*\}$ where number of 0's and 1's in w are divisible by 3 and 5, respectively, has a) 15 states ✓ b) 11 states c) 10 states d) 9 states	CORRECT ANSWER Answer: A 15 states
15. Following are two finite automata Y and Z ($\rightarrow$ indicates the start state and F indicates the final state). Which of the following represents the product automaton $Z \times Y$? a) Table (a) ✓ b) Table (b) ✓ c) Table (c) d) Table (d)	CORRECT ANSWER Answer: A, B Table (a) Table (b)
16. The given DFA accepts the set of all strings over $\{0, 1\}$ that a) begin either with 0 or 1. b) end with 0 c) end with 00. ✓ d) contain the substring 00.	CORRECT ANSWER Answer: C end with 00.
17. Let w be any string of length n in $\{0, 1\}^*$. Let L be the set of all substrings of w. What is the minimum number of states in a non-deterministic FA that accepts L? a) $n - 1$ b) n c) $n + 1$ ✓ d) $2^n - 1$	CORRECT ANSWER Answer: C $n + 1$
18. Definition of a language L with alphabet $\{a\}$ is given as following $\{a^{nk} \mid k > 0 \text{ and } n \text{ is a positive integer constant}\}$. What is the minimum number of states needed in a DFA to recognize L? a) $k + 1$ b) $n + 1$ ✓ c) 2^{n+1} d) 2^{k+1}	CORRECT ANSWER Answer: B $n + 1$
19. A deterministic finite automation (DFA) D with alphabet $\Sigma = \{a, b\}$ is given. Which of the following finite state machines is a valid minimal DFA which accepts the same language as D? a) Machine (a) ✓ b) Machine (b) c) Machine (c) d) Machine (d)	CORRECT ANSWER Answer: A Machine (a)
20. The _____ of a set of states, P, of an NFA is defined as the set of states reachable from any state in P following e-transitions. a) e-closure ✓ b) e-pack c) Q in the tuple d) None of the mentioned	CORRECT ANSWER Answer: A e-closure
21. The transition function of NFA is a) $QX\Sigma \rightarrow Q$ b) $QX\Sigma \rightarrow 2^n$ c) $QX\Sigma \rightarrow Q^n$ d) $QX\Sigma \rightarrow 2^Q$ ✓	CORRECT ANSWER Answer: D $QX\Sigma \rightarrow 2^Q$

1. The machine format of regular expression is a) Finite automata ✓ b) Push down automata c) Turing machine d) All of the above	CORRECT ANSWER Answer: A Finite automata
2. The regular expression is accepted by a) Finite automata b) Push down automata c) Turing machine d) All of the above ✓	CORRECT ANSWER Answer: D All of the above
3. The language of all words with at least 2 a's can be described by the regular expression a) (ab)*a b) (a + b)*ab*a(a + b)* ✓ c) b*ab*a(a + b)* d) All of the above	CORRECT ANSWER Answer: B (a + b)*ab*a(a + b)*
4. The set of all strings of {0, 1} having exactly two 0's is a) 1*01*01* ✓ b) {0 + 1}*1 c) {11 + 0}* d) {00 + 11}*	CORRECT ANSWER Answer: A 1*01*01*
5. The set of all strings of {0, 1} where 0 is followed by 1 and 1 is followed by 0 is a) (01)* b) 0*1* c) 0*1 + 1*0 d) 0*1* + 1*0* ✓	CORRECT ANSWER Answer: D 0*1* + 1*0*
6. Which of the strings do not belong to the regular expression (ba + baa)*aaba a) baaaba b) babaabaaaba c) bababababa ✓ d) baaaaba	CORRECT ANSWER Answer: C bababababa
7. Which of the following type of language is regular expression? a) Type 0 b) Type 1 c) Type 2 ✓ d) Type 3	CORRECT ANSWER Answer: C Type 2
8. Which of the following RE over {0,1} denotes the set of all strings not containing 100 as substring? a) (1 + 0)*0* b) 0*1010* c) 0*1*01 ✓ d) All of the above	CORRECT ANSWER Answer: C 0*1*01
9. $\epsilon + RR^* = ?$ a) R b) R* ✓ c) R^+ d) ϵ	CORRECT ANSWER Answer: B R*
10. {a/b} denotes the set a) {ab} b) {a, b} ✓ c) {a, b}* d) {ab}*	CORRECT ANSWER Answer: B {a, b}
11. What is the solution for the equation $R = Q + RP$ (If P and Q are RE and P does not contain Λ)? a) R = QP* ✓ b) $R = QP$ c) $R = PQ^*$ d) $R = P^*Q^*$	CORRECT ANSWER Answer: A $R = QP^*$

12. Which of the languages is accepted by the following FA?

- a) **$b(a + bba^*)^*a^*$** ✓
- b) $a(a + bba^*)^*b^*$
- c) $a(a + bb^*a^*)^*b^*$
- d) None of these

CORRECT ANSWER

Answer: A

$b(a + bba^*)^*a^*$

13. Which is true for ϵ -closure of a state?

- a) All the states reachable from the state with input null excluding the state
- b) The state only
- c) All the other states
- d) **All the states reachable from the state with input null** ✓

CORRECT ANSWER

Answer: D

All the states reachable from the state with input null

14. The pumping lemma for regular expression is used to prove that

- a) **Certain sets are regular** ✓
- b) Certain sets are not regular
- c) Certain regular grammar produce RE
- d) Certain regular grammar does not produce RE

CORRECT ANSWER

Answer: A

Certain sets are regular

15. Regular sets are closed under

- a) Union
- b) Concatenation
- c) Kleene closure
- d) **All of the above** ✓

CORRECT ANSWER

Answer: D

All of the above

16. Which among the following looks similar to the given expression? $((0+1).(0+1))^*$

- a) **$\{x\epsilon\{0,1\}^* \mid x \text{ is all binary number with even length}\}$** ✓
- b) $\{x\epsilon\{0,1\} \mid x \text{ is all binary number with even length}\}$
- c) $\{x\epsilon\{0,1\}^* \mid x \text{ is all binary number with odd length}\}$
- d) $\{x\epsilon\{0,1\} \mid x \text{ is all binary number with odd length}\}$

CORRECT ANSWER

Answer: A

$\{x\epsilon\{0,1\}^* \mid x \text{ is all binary number with even length}\}$

GATE Questions — Regular Expressions

1. Which two of the following four regular expressions are equivalent? (i) $(00)^*$ ($\epsilon + 0$) (ii) $(00)^*$ (iii) 0^*0^* (iv) $0(00)^*$

- a) (i) and (iii)
- b) **(i) and (iv)** ✓
- c) (ii) and (iii)
- d) (iii) and (iv)

CORRECT ANSWER

Answer: B

(i) and (iv)

2. Which of the following alternatives is true for the following three regular expressions? (i) $011((11)^* + (01)^*)^*011$ (ii) $011(((1 + 0)1)^*011)^*$ (iii) $011(((1 + 0)^*1)^*011)^*$

- a) (i) and (iii) are equivalent
- b) (ii) and (iii) are equivalent
- c) **(i) and (ii) are equivalent** ✓
- d) No pairs are equivalent

CORRECT ANSWER

Answer: C

(i) and (ii) are equivalent

3. Consider the following statements: (i) $\emptyset^* = \epsilon$ (ii) $s(rs + s)^*r = (sr + s)^*sr$ (iii) $(r + s)^* = (r^* + s^*)^+$. Find out the correct alternatives.

- a) All are true
- b) Only (ii) is false
- c) only (iii) is false
- d) **Both (ii) and (iii) are false** ✓

CORRECT ANSWER

Answer: D

Both (ii) and (iii) are false

4. Which regular expression best describes the language accepted by the following non-deterministic automation?

- a) **$(a + b)^*a(a + b)b$** ✓
- b) $\{abb\}^*$
- c) $(a + b)^*a(a + b)^*b(a + b)^*$
- d) $(a + b)^*$

CORRECT ANSWER

Answer: A

$(a + b)^*a(a + b)b$

5. Which of the following regular expressions describe the language over $\{0, 1\}$ consisting of strings that contain exactly two 1's?

- a) $(0 + 1)^*11(0 + 1)^*$
- b) 0^*110^*
- c) **$0^*10^*10^*$** ✓
- d) $(0 + 1)^*1(0 + 1)^*1(0 + 1)^*$

CORRECT ANSWER

Answer: C

$0^*10^*10^*$

6. Find the false statement if $S = (a, b)$ a) $L = \{a^n b^m, m, n \geq 1\}$ is regular b) $L = \{X, \text{ where no of } a > \text{no of } b\}$ is not regular c) $L = \{a^n b^n, n \geq 1\}$ is regular ✓ d) $L = \{X, \text{ where no of } a \text{ and no of } b \text{ are equal}\}$ is not regular	CORRECT ANSWER Answer: C $L = \{a^n b^n, n \geq 1\}$ is regular
7. Which of the following regular expression identified is true? a) $r^{(*)} = r^*$ ✓ b) $(r^*s^*) = (r + s)^*$ c) $(r + s)^* = r^* + s^*$ d) $r^*s^* = r^* + s^*$	CORRECT ANSWER Answer: A $r^{(*)} = r^*$
8. In some programming languages, an identifier is permitted to be a letter followed by any number of letters or digits. If L and D denote the sets of letters and digits, respectively, which of the following expressions define an identifier? a) $(L \cup D)^*$ b) $L(L \cup D)^*$ ✓ c) $(L.D)^*$ d) $L(L.D)^*$	CORRECT ANSWER Answer: B $L(L \cup D)^*$
9. Which two of the following four regular expressions are equivalent? (i) $(00)^*$ ($\in + 0$) (ii) $(00)^*$ (iii) 0^* (iv) $0(00)^*$ a) (i) and (ii) b) (ii) and (iii) c) (i) and (iii) ✓ d) (iii) and (iv)	CORRECT ANSWER Answer: C (i) and (iii)
10. Let $L \subseteq \Sigma^*$, where $\Sigma = \{a, b\}$. Which of the following is true? a) $L = \{x \mid x \text{ has equal number of } a\text{'s and } b\text{'s}\}$ is regular b) $L = \{a^n b^n \mid n \geq 1\}$ is regular c) $L = \{x \mid x \text{ has more } a\text{'s than } b\text{'s}\}$ is regular d) $L = \{a^m b^n \mid m \geq 1, n \geq 1\}$ is regular ✓	CORRECT ANSWER Answer: D $L = \{a^m b^n \mid m \geq 1, n \geq 1\}$ is regular
11. The string 1101 does not belong to the set represented by a) $110^*(0 + 1)$ b) $1(0 + 1)^*101$ c) $(10)^*(01)^*(00 + 11)^*$ ✓ d) $(00 + (11)^*01)^*$	CORRECT ANSWER Answer: C $(10)^*(01)^*(00 + 11)^*$
12. Give a regular expression for the set of binary strings where every 0 is immediately followed by exactly k number of 1's and preceded by at least k number of 1's (k is a fixed integer). a) $1^*1^k(01^k)^*1^*$ ✓ b) $1^*1^k01^k$ c) $1^*(1^k01^k)^*$ d) $1^*(1^k0)^*1^*$	CORRECT ANSWER Answer: A $1^*1^k(01^k)^*1^*$
13. Let S and T be the languages over $\Sigma = \{a, b\}$ represented by the regular expressions $(a + b^*)^*$ and $(a + b)^*$, respectively. Which of the following is true? a) $S \subset T$ b) $T \subset S$ c) $S = T$ ✓ d) $S \cap T = \emptyset$	CORRECT ANSWER Answer: C $S = T$
14. The regular expression $0^*(10^*)^*$ denotes the same set as a) $(1^*0)^*1^*$ ✓ b) $0 + (0 + 10)^*$ c) $(0 + 1)^*10(0 + 1)^*$ d) None of the above	CORRECT ANSWER Answer: A $(1^*0)^*1^*$
15. Consider the following NFA M. Let the language accepted by M be L. Let L_1 be the language accepted by the NFA M_1 obtained by changing the accepting states of M to a non-accepting state and by changing the non-accepting states of M to accepting states. Which of the following statements is true? a) $L_1 = \{0, 1\}^* - L$ b) $L_1 = \{0, 1\}^*$ ✓ c) $L_1 \subseteq L$ d) $L_1 = L$	CORRECT ANSWER Answer: B $L_1 = \{0, 1\}^*$

16. (Diagram-based DFA question — strings over {0,1})

- a) Option a
- b) Option b ✓
- c) Option c
- d) Option d

CORRECT ANSWER

Answer: B
Option b

17. Consider the regular language $L = (111 + 11111)^*$. The minimum number of states in any DFA accepting this language is

- a) 3
- b) 5
- c) 8 ✓
- d) 9

CORRECT ANSWER

Answer: C
8

18. Which of the following is true?

- a) Every subset of a regular set is regular.
- b) Every finite subset of a non-regular set is regular.
- c) The union of two non-regular sets is not regular. ✓
- d) Infinite union of finite sets is regular.

CORRECT ANSWER

Answer: C
The union of two non-regular sets is not regular.

19. The language accepted by this automaton is given by the regular expression

- a) $b^*ab^*ab^*ab^*$
- b) $(a + b)^*$ ✓
- c) $b^*a(a + b)^*$
- d) $b^*ab^*ab^*$

CORRECT ANSWER

Answer: B
 $(a + b)^*$

20. The minimum state automation equivalent to the previous FSA has the following number of states.

- a) 1
- b) 2
- c) 3
- d) 4 ✓

CORRECT ANSWER

Answer: D
4

21. Which of the following is true for the language $\{a^p \mid p \text{ is prime}\}$?

- a) It is not accepted by a Turing machine
- b) It is regular but not context-free
- c) It is context-free but not regular ✓
- d) It is neither regular nor context-free but accepted by a Turing machine.

CORRECT ANSWER

Answer: C
It is context-free but not regular

22. Match the following NFAs (P, Q, R, S) with the regular expressions they correspond to: 1. $\epsilon + 0(01^*1 + 00)^*01^*$ 2. $\epsilon + 0(10^*1 + 00)^*0$ 3. $\epsilon + 0(10^*1 + 10)^*1$ 4. $\epsilon + 0(10^*1 + 10)^*10^*$

- a) P–2, Q–1, R–3, S–4
- b) P–1, Q–3, R–2, S–4
- c) P–1, Q–2, R–3, S–4 ✓
- d) P–3, Q–2, R–1, S–4

CORRECT ANSWER

Answer: C
P–1, Q–2, R–3, S–4

23. Which of the following are regular sets? I. $\{a^n b^{2m} \mid n \geq 0, m \geq 0\}$ II. $\{a^n b^m \mid n = 2m\}$ III. $\{a^n b^m \mid n \neq m\}$ IV. $\{xyc \mid x, y \in \{a, b\}^*\}$

- a) I and IV only
- b) I and III only
- c) I only ✓
- d) IV only

CORRECT ANSWER

Answer: C
I only

24. Which one of the following languages over the alphabet $\{0, 1\}$ is described by the regular expression: $(0 + 1)^*0(0 + 1)^*0(0 + 1)^*$?

- a) The set of all strings containing the substring 00. ✓
- b) The set of all strings containing at most two 0's. ✓
- c) The set of all strings containing at least two 0's.
- d) The set of all strings that begin and end with either 0 or 1 ✓

CORRECT ANSWER

Answer: A, B, D
The set of all strings containing the substring 00.
The set of all strings containing at most two 0's.
The set of all strings that begin and end with either 0 or 1

25. Let $L = \{w \in (0 + 1)^* \mid w \text{ has an even number of 1s}\}$, i.e., L is the set of all bit strings with even number of 1s. Which one of the given regular expressions represent L?

- a) $(0^*10^*1)^*$
- b) $0^*(10^*10^*)^*$ ✓
- c) $0^*(10^*1^*)^*0^*$
- d) $0^*1(10^*1)^*10^*$

CORRECT ANSWER

Answer: B
 $0^*(10^*10^*)^*$

26. What is the complement of the language accepted by the given NFA? Assume that $\Sigma = \{a\}$ and ϵ is the empty string.

- a) ϵ
- b) $\{\epsilon\}$
- c) **a^*** ✓
- d) $\{a, \epsilon\}$

CORRECT ANSWER

Answer: C

a^*

27. Given the language $L = \{ab, aa, baa\}$, which of the following strings are in L^* ? 1. abaabaaabaa 2. aaaabaaaa 3. baaaaabaaaab 4. baaaaabaa

- a) 1, 2, and 3
- b) 2, 3, and 4
- c) **1, 2, and 4** ✓
- d) 1, 3, and 4

CORRECT ANSWER

Answer: C

1, 2, and 4

28. Which one of the following regular expressions is not equivalent to the regular expression $(a + b + c)^*$?

- a) $(a^* + b^* + c^*)^*$
- b) **$(a^*b^*c^*)^*$** ✓
- c) $((ab)^* + c^*)^*$
- d) $(a^*b^* + c^*)$

CORRECT ANSWER

Answer: B

$(a^*b^*c^*)^*$

UNIT - II: Context-Free Grammars — Multiple Choice Questions

1. Context-free language is _____ language.

- a) Type 0
- b) Type 1
- c) **Type 2** ✓
- d) Type 3

CORRECT ANSWER

Answer: C

Type 2

2. Parsing a string from a given grammar means

- a) **Finding a derivation** ✓
- b) Finding a leftmost derivation
- c) Finding a rightmost derivation
- d) Finding a derivation tree.

CORRECT ANSWER

Answer: A

Finding a derivation

3. A grammar is called ambiguous if

- a) It generates more than one string
- b) It generates both leftmost and rightmost derivation for a given string
- c) **It generates more than one parse tree for a given string** ✓
- d) It fulfills both (b) and (c)

CORRECT ANSWER

Answer: C

It generates more than one parse tree for a given string

4. Which is not true for ambiguous grammar?

- a) Ambiguity creates problem in generating a language from a given grammar
- b) **All ambiguity can be removed.** ✓
- c) Inherent ambiguity cannot be removed
- d) Some ambiguity can be removed by hand

CORRECT ANSWER

Answer: B

All ambiguity can be removed.

5. Non-generating symbols are those symbols which

- a) Do not generate any string of non-terminals
- b) Do not generate any null string
- c) Do not generate any string of terminal and non-terminals
- d) **Do not generate any string of terminals** ✓

CORRECT ANSWER

Answer: D

Do not generate any string of terminals

6. Useless symbols in CFG are

- a) **Non-generating symbols and non-reachable symbols** ✓
- b) Null alphabets and null string
- c) Non-terminal symbols
- d) All of these

CORRECT ANSWER

Answer: A

Non-generating symbols and non-reachable symbols

7. Which of the following is a unit production?

- a) $(\text{String of NT}) \rightarrow (\text{String of NT})$
- b) $(\text{Single NT}) \rightarrow (\text{String of NT})$
- c) **$(\text{Single NT}) \rightarrow (\text{Single NT})$** ✓
- d) $(\text{String of NT}) \rightarrow (\text{Single NT})$

CORRECT ANSWER

Answer: C

$(\text{Single NT}) \rightarrow (\text{Single NT})$

8. Which is true for the following CFG? $S \rightarrow aA/\epsilon$, $A \rightarrow bA/a$

- a) Null production can be removed
- b) **Null production cannot be removed** ✓
- c) As A does not produce null, null cannot be removed
- d) Both (b) and (c)

CORRECT ANSWER

Answer: B

Null production cannot be removed

9. Which of the following production is in CNF? (more specific)

- a) $(NT) \rightarrow (String\ of\ NT)$
- b) $(NT) \rightarrow (String\ of\ terminal\ and\ non-terminal)$
- c) $(NT) \rightarrow (String\ of\ terminal)$
- d) **$(NT) \rightarrow (String\ of\ exactly\ two\ NT)$** ✓

CORRECT ANSWER

Answer: D

$(NT) \rightarrow (String\ of\ exactly\ two\ NT)$

10. Which of the following production is in GNF? (more specific)

- a) **$(NT) \rightarrow (Single\ T)(String\ of\ NT)$** ✓
- b) $(NT) \rightarrow (Single\ NT)(String\ of\ T)$
- c) $(NT) \rightarrow (String\ of\ terminal\ and\ non-terminal)$
- d) $(NT) \rightarrow (String\ of\ NT)$

CORRECT ANSWER

Answer: A

$(NT) \rightarrow (Single\ T)(String\ of\ NT)$

11. Which of the following is common in both CNF and GNF?

- a) $(NT) \rightarrow (Single\ T)(String\ of\ NT)$
- b) $(NT) \rightarrow (String\ of\ exactly\ two\ NT)$
- c) $(NT) \rightarrow (String\ of\ NT)$
- d) **$(NT) \rightarrow (Single\ T)$** ✓

CORRECT ANSWER

Answer: D

$(NT) \rightarrow (Single\ T)$

12. Context-free language is not closed under

- a) Union
- b) Concatenation
- c) **Complementation** ✓
- d) Star closure

CORRECT ANSWER

Answer: C

Complementation

CFG & Properties of CFL

1. The intersection of CFL and RE is always

- a) **CFL** ✓
- b) RE
- c) CSL
- d) CFL or CSL

CORRECT ANSWER

Answer: A

CFL

2. Consider an ambiguous grammar G and its disambiguated version D. Let the language recognized by the two grammar be denoted by L(G) and L(D), respectively. Which of the following is true?

- a) **$L(D) \subset L(G)$** ✓
- b) $L(D) \supset L(G)$
- c) $L(D) = L(G)$
- d) L(D) is empty

CORRECT ANSWER

Answer: A

$L(D) \subset L(G)$

3. Consider a CFG with the productions $S \rightarrow AA/B$, $A \rightarrow 0A/A0/1$, $B \rightarrow 0B00/1$. S is the start symbol, A and B are non-terminals, and 0 and 1 are the terminals. The language generated by the grammar is

- a) $\{0^n10^{2n} \mid n \geq 1\}$
- b) $\{0^i10^j10^k \mid i, j, k \geq 0\} \cup \{0^n10^{2n} \mid n \geq 1\}$
- c) **$\{0^i10^j \mid i, j \geq 0\} \cup \{0^n10^{2n} \mid n \geq 1\}$** ✓
- d) The set of all strings over $\{0, 1\}$ containing at least two 0's

CORRECT ANSWER

Answer: C

$\{0^i10^j \mid i, j \geq 0\} \cup \{0^n10^{2n} \mid n \geq 1\}$

4. If G is a CFG and w is a string of length l in L(G), how long is a derivation of w in G, if G is a Chomsky normal form?

- a) 2l
- b) $2l + 1$
- c) **$2l - 1$** ✓
- d) l

CORRECT ANSWER

Answer: C

$2l - 1$

5. Context-free languages are

- a) **closed under union** ✓
- b) closed under complementation
- c) closed under intersection
- d) **closed under Kleene closure** ✓

CORRECT ANSWER

Answer: A, D

closed under union
closed under Kleene closure

6. Which of the following features cannot be captured by context-free grammar?

- a) Syntax of if-then-else statements
- b) **Syntax of recursive procedures** ✓
- c) Whether a variable has been declared before use
- d) Variable names of arbitrary length

CORRECT ANSWER

Answer: B

Syntax of recursive procedures

7. Let $\Sigma = \{0, 1\}$, $L = \Sigma^*$ and $R = \{0^n 1^n \text{ such that } n > 0\}$, then the language $L \cup R$ and R are respectively.

- a) regular, regular
- b) not regular, regular
- c) regular, not regular
- d) **not regular, not regular** ✓

CORRECT ANSWER

Answer: D

not regular, not regular

8. The grammar whose productions are $\langle \text{Start} \rangle \rightarrow \text{if id then } \langle \text{stmt} \rangle \mid \text{if id then } \langle \text{stmt} \rangle \text{ else } \langle \text{stmt} \rangle \mid \text{id} := \text{id}$, is ambiguous because

- a) the sentence 'if a then if b then c := d'
- b) the leftmost and rightmost derivations of the sentence 'if a then if b then c:=d' give rise to two different parse trees
- c) the sentence 'if a then if b then c:=d else c:=f' has more than two parse trees
- d) **the sentence 'if a then if b then c:=d else c:=f' has two parse trees** ✓

CORRECT ANSWER

Answer: D

the sentence 'if a then if b then c:=d else c:=f' has two parse trees

9. If L_1 is a context-free language and L_2 is a regular language, which of the following is/are false?

- a) $L_1 - L_2$ is not context free
- b) $L_1 \cap L_2$ is context free
- c) **$\sim L_1$ is context free** ✓
- d) $\sim L_2$ is regular

CORRECT ANSWER

Answer: C

$\sim L_1$ is context free

10. A grammar that is both left and right recursive for a non-terminal, is

- a) Ambiguous
- b) Unambiguous
- c) **Information is not sufficient to decide whether it is ambiguous or unambiguous.** ✓
- d) None of the above

CORRECT ANSWER

Answer: C

Information is not sufficient to decide whether it is ambiguous or unambiguous.

11. Let L denote the language generated by the grammar $S \rightarrow 0S0/00$. Which of the following is true?

- a) $L = 0^+$
- b) **L is regular but not 0^+** ✓
- c) L is context free but not regular
- d) L is not context free

CORRECT ANSWER

Answer: B

L is regular but not 0^+

12. Consider the following two statements: $S_1: \{0^{2^n} \mid n \geq 1\}$ is a regular language; $S_2: \{0^m 1^n 0^{m+n} \mid m \geq 1 \text{ and } n \geq 1\}$ is a regular language. Which of the following statements is correct?

- a) **Only S_1 is correct** ✓
- b) Only S_2 is correct
- c) Both S_1 and S_2 are correct
- d) None of S_1 and S_2 is correct

CORRECT ANSWER

Answer: A

Only S_1 is correct

13. Consider the following languages: $L_1 = \{ww \mid w \in \{a, b\}^*\}$, $L_2 = \{ww^R \mid w \in \{a, b\}^* \text{ and } w^R \text{ is the reverse of } w\}$, $L_3 = \{0^{2^i} \mid i \text{ is an integer}\}$, $L_4 = \{0^{i^2} \mid i \text{ is an integer}\}$. Which of the languages are regular?

- a) Only L_1 and L_2
- b) Only L_2 , L_3 , and L_4
- c) Only L_3 and L_4
- d) **Only L_3** ✓

CORRECT ANSWER

Answer: D

Only L_3

14. Which of the following grammar rules violate the requirements of an operator grammar? P, Q , and R are non-terminals, and r, s , and t are terminals. (i) $P \rightarrow QR$ (ii) $P \rightarrow QsR$ (iii) $P \rightarrow \epsilon$ (iv) $P \rightarrow QtRr$

- a) (i) only
- b) **(i) and (iii) only** ✓
- c) (ii) and (iii) only
- d) (iii) and (iv) only

CORRECT ANSWER

Answer: B

(i) and (iii) only

15. Consider the languages: $L_1 = \{a^n b^n c^m \mid n, m > 0\}$ and $L_2 = \{a^n b^m c^m \mid n, m > 0\}$. Which one of the following statements is false?

- a) **$L_1 \cap L_2$ is a context-free language** ✓
- b) $L_1 \cup L_2$ is a context-free language
- c) L_1 and L_2 are context-free languages
- d) $L_1 \cap L_2$ is a context-sensitive language

CORRECT ANSWER

Answer: A

$L_1 \cap L_2$ is a context-free language

16. Consider the following statements about the CFG $G = \{S \rightarrow SS, S \rightarrow ab, S \rightarrow ba, S \rightarrow \epsilon\}$: (i) G is ambiguous (ii) G produces all strings with equal number of a 's and b 's (iii) G can be accepted by a deterministic PDA. Which of the following combination expresses all the true statements about G ?

- a) (i) only
- b) (i) and (iii) only
- c) (ii) and (iii) only
- d) (i), (ii), and (iii) ✓

CORRECT ANSWER

Answer: D

(i), (ii), and (iii)

17. Consider the grammar $S \rightarrow aB/bA/bAA, A \rightarrow a/aS, B \rightarrow b/bS/aBB$. Which of the following strings is generated by the grammar?

- a) aaaabb
- b) aabbbb
- c) aabbab ✓
- d) abbbba

CORRECT ANSWER

Answer: C

aabbab

18. For the correct answer of the previous question, how many derivation trees are there?

- a) 1
- b) 2 ✓
- c) 3
- d) 4

CORRECT ANSWER

Answer: B

2

19. Which of the following statements are true? (i) Every left-recursive grammar can be converted to a right-recursive grammar and vice versa. (ii) All $\in$ productions can be removed from any context-free grammar by suitable transformations. (iii) The language generated by a CFG, all of whose productions are of the form $X \rightarrow w$ or $X \rightarrow wY$ (where w is a string of terminals and Y is a non-terminal), is always regular. (iv) The derivation trees of strings generated by a CFG in CNF are always binary trees.

- a) (i), (ii), (iii), and (iv)
- b) (ii), (iii), and (iv) only
- c) (i), (iii) and (iv) only ✓
- d) (i), (ii) and (iv) only

CORRECT ANSWER

Answer: C

(i), (iii) and (iv) only

20. Match the following: E-Checking that identifiers are declared before their use; F-The number of formal parameters agrees with the number of actual parameters; G-Arithmetic expression with matched pairs of parentheses; H-Palindrome. P: $L = \{a^n b^m c^n d^m \mid n \geq 1, m \geq 1\}$; Q: $X \rightarrow XbX \mid XcX \mid dXf \mid g$; R: $L = \{wcw \mid w \in (a|b)^*\}$; S: $X \rightarrow bXb \mid cXc \mid \epsilon$

- a) E-P, F-R, G-Q, H-S
- b) E-R, F-P, G-S, H-Q
- c) E-R, F-P, G-Q, H-S ✓
- d) E-P, F-R, G-S, H-Q

CORRECT ANSWER

Answer: C

E-R, F-P, G-Q, H-S

21. Which one of the following statements is false?

- a) There exist context-free languages such that all the CFGs generating them are ambiguous. ✓
- b) An unambiguous CFG always has a unique parse tree for each string of the language generated by it.
- c) Both deterministic and non-deterministic push down automata always accept the same set of languages.
- d) A finite set of string from one alphabet is always a regular language.

CORRECT ANSWER

Answer: A

There exist context-free languages such that all the CFGs generating them are ambiguous.

GATE Questions — Context-Free Grammars

1. In the given context-free grammar, S is the start symbol, a and b are terminals, and ϵ denotes the empty string. $S \rightarrow aSa/bSb/a/b/\epsilon$. Which of the following strings is NOT generated by the grammar?

- a) aaaa
- b) baba ✓
- c) abba
- d) babaaabab

CORRECT ANSWER

Answer: B

baba

2. In the given context-free grammar, $S \rightarrow aSAb/\epsilon, A \rightarrow bA/\epsilon$. The grammar generates the language

- a) $((a + b)^*b)^*$
- b) $\{a^m b^n \mid m \leq n\}$ ✓
- c) $\{a^m b^n \mid m = n\}$
- d) a^*b^*

CORRECT ANSWER

Answer: B

$\{a^m b^n \mid m \leq n\}$

3. The two grammars given generate a language over the alphabet (x, y, z).
G1: $S \rightarrow x|z|xS|zS|yB$, $B \rightarrow y|z|yB|zB$; G2: $S \rightarrow y|z|yS|zS|xB$, $B \rightarrow y|yS$. Which of the following choices describe the properties satisfied by the strings in these languages?

- a) **G1: No y appears before any x; G2: Every x is followed by at least one y** ✓
b) G1: No y appears before any x; G2: No x appears before any y
c) G1: No y appears after any x; G2: Every x is followed by at least one y.
d) G1: No y appears after any x; G2: Every y is followed by at least one x.

4. Consider the given grammar $S \rightarrow xB/yA$, $A \rightarrow x/xS/yAA$, $B \rightarrow y/yS/xBB$. Consider the following strings. Which are accepted by the grammar? i) xxyxy ii) xxyxyx iii) xyxy iv) yxxy v) yxx vi) yxx

- a) (i), (ii) and (iii)
b) (ii), (v) and (vi)
c) **(ii), (iii) and (iv)** ✓
d) (i), (iii) and (iv)

5. Consider the following grammars G1 and G2 (names representing terminals given in capital letters): G1: $stmt \rightarrow WHILE (expr) stmt$, $stmt \rightarrow OTHER$, $expr \rightarrow ID$; G2: $adds\ expr \rightarrow expr + expr \mid expr * expr \mid ID$. Which of the following statements is true?

- a) G1 is context-free but not regular and G2 is regular
b) **G2 is context-free but not regular and G1 is regular** ✓
c) Both G1 and G2 are regular
d) Both G1 and G2 are context-free but neither of them is regular.

6. Which of the following definitions generate the same language as L, where $L = \{x^n y^n \mid \text{such that } n \geq 1\}$? I. $E \rightarrow xEy/xy$ II. $xy/(x^+ xyy^+)$ III. $x^+ y^+$

- a) **I only** ✓
b) I and II
c) II and III
d) II only

7. Consider the grammar $S \rightarrow bSe/PQR$, $P \rightarrow bPc/\epsilon$, $Q \rightarrow cQd/\epsilon$, $R \rightarrow dRe/\epsilon$. The string generated by the grammar is $b^i c^j d^k e^m$ for some i, j, k, m ≥ 0. What is the relation among i, j, k, and m?

- a) $i = j = k = m$
b) **$i + k = j + m$** ✓
c) $i = j$ but $k \neq m$
d) $i + j + k = m$

8. Consider the grammar $S \rightarrow aSAb/\epsilon$, $A \rightarrow bA/\epsilon$. The grammar generates the strings of the form $a^i b^j$ for some i, j ≥ 0. What are the conditions of the values of i and j?

- a) $i = j$
b) $j \leq 2i$
c) $j \geq 2i$
d) **$i \leq j$** ✓

9. The language $\{a^m b^n c^{m+n} \mid m, n \geq 1\}$ is

- a) regular
b) **context-free but not regular** ✓
c) context-sensitive but not context-free
d) type-0 but not context-sensitive

10. Consider the following grammar G: $S \rightarrow bS/aA/b$, $A \rightarrow bA/aB$, $B \rightarrow bB/aS/a$. Let $N_a(w)$ and $N_b(w)$ denote the number of a's and b's in a string w, respectively. The language $L(G) \subseteq \{a, b\}^+$ generated by G is

- a) $\{w \mid N_a(w) > 3N_b(w)\}$
b) $\{w \mid N_b(w) > 3N_a(w)\}$
c) **$\{w \mid N_a(w) = 3k, k \in \{0, 1, 2, \dots\}\}$** ✓
d) $\{w \mid N_b(w) = 3k, k \in \{0, 1, 2, \dots\}\}$

11. Let $L_1 = \{0^{n+m} 1^n 0^m \mid n, m \geq 0\}$, $L_2 = \{0^{n+m} 1^{n+m} 0^m \mid n, m \geq 0\}$, and $L_3 = \{0^{n+m} 1^{n+m} 0^{n+m} \mid n, m \geq 0\}$. Which of these languages are NOT context-free?

- a) L_1 only
b) L_3 only
c) L_1 and L_2
d) **L_2 and L_3** ✓

CORRECT ANSWER

Answer: A

G1: No y appears before any x; G2: Every x is followed by at least one y

CORRECT ANSWER

Answer: C

(ii), (iii) and (iv)

CORRECT ANSWER

Answer: B

G2 is context-free but not regular and G1 is regular

CORRECT ANSWER

Answer: A

I only

CORRECT ANSWER

Answer: B

$i + k = j + m$

CORRECT ANSWER

Answer: D

$i \leq j$

CORRECT ANSWER

Answer: B

context-free but not regular

CORRECT ANSWER

Answer: C

$\{w \mid N_a(w) = 3k, k \in \{0, 1, 2, \dots\}\}$

CORRECT ANSWER

Answer: D

L_2 and L_3

12. Which one of the following grammars generate the language $L = \{a^i b^j \mid i \neq j\}$?

a) Grammar (a)
b) $S \rightarrow aS \mid Sb \mid a \mid b$
c) Grammar (c)
d) **Grammar (d)** ✓

CORRECT ANSWER

Answer: D

Grammar (d)

13. In the correct grammar above, what is the length of the derivation (number of steps starting from S) to generate the string $a^l b^m$ with $l \neq m$?

a) **$\max(l, m) + 2$** ✓
b) $l + m + 2$
c) $l + m + 3$
d) $\max(l, m) + 3$

CORRECT ANSWER

Answer: A

$\max(l, m) + 2$

14. $S \rightarrow aSa \mid bSb \mid a \mid b$. The language generated by the above grammar over the alphabet $\{a, b\}$ is the set of

a) all palindromes.
b) **all odd length palindromes.** ✓
c) strings that begin and end with the same symbol.
d) all even length palindromes.

CORRECT ANSWER

Answer: B

all odd length palindromes.

15. Consider the languages $L_1 = \{0^i 1^j \mid i \neq j\}$, $L_2 = \{0^i 1^j \mid i = j\}$, $L_3 = \{0^i 1^j \mid i = 2j + 1\}$, and $L_4 = \{0^i 1^j \mid i \neq 2j\}$. Which one of the following statements is true?

a) Only L_2 is context-free
b) Only L_2 and L_3 are context-free
c) Only L_1 and L_2 are context-free
d) **All are context-free** ✓

CORRECT ANSWER

Answer: D

All are context-free

UNIT - II: Pushdown Automata — Multiple Choice Questions

1. PDA is the machine format of

a) Type 0 language
b) Type 1 language
c) **Type 2 language** ✓
d) Type 3 language

CORRECT ANSWER

Answer: C

Type 2 language

2. Which is not true for mechanical diagram of PDA?

a) PDA contains a stack
b) **The head reads as well as writes** ✓
c) The head moves from left to right
d) The input string is surrounded by an infinite number of blanks in both sides

CORRECT ANSWER

Answer: B

The head reads as well as writes

3. The difference between finite automata and pushdown automata is in _____.

a) Reading head
b) Input tape
c) Finite control
d) **Stack** ✓

CORRECT ANSWER

Answer: D

Stack

4. In the PDA, transitional function δ is in the form (Q finite states, Σ input alphabets, Γ stack symbols)

a) **$Q \times (\Sigma \cup \{\lambda\}) \times \Gamma \rightarrow (Q, \Gamma)$** ✓
b) $Q \times \Sigma \rightarrow Q$
c) $Q \times \Sigma \times \Gamma \rightarrow Q$
d) $Q \times \Gamma \rightarrow Q \times \Sigma$

CORRECT ANSWER

Answer: A

$Q \times (\Sigma \cup \{\lambda\}) \times \Gamma \rightarrow (Q, \Gamma)$

5. Instantaneous description remembers

a) The information of state and input tape content at a given instance of time.
b) **The information of state and stack content at a given instance of time.** ✓
c) The information of input tape and stack content at a given instance of time.
d) The information of state, input tape, and stack content at a given instance of time.

CORRECT ANSWER

Answer: B

The information of state and stack content at a given instance of time.

6. Which of the following is not possible algorithmically?

a) RE to CFG
b) NFA to DFA
c) CFG to PDA
d) **NPDA to DPDA** ✓

CORRECT ANSWER

Answer: D

NPDA to DPDA

GATE Questions — Pushdown Automata

1. Which of the following is not accepted by a DPDA but accepted by an NPDA?

- a) $L = a^n b^n$, where $n > 0$
- b) $L = WCW^R$, where $W \in (a, b)^*$
- c) **$L = WW^R$, where $W \in (a, b)^+$** ✓
- d) $L = a^n b^m c^m d^n$, where $m, n > 0$

CORRECT ANSWER

Answer: C

$L = WW^R$, where $W \in (a, b)^+$

2. Which of the followings cannot be designed by a PDA?

- a) $a^n b^n c^i$, where $n, i > 0$
- b) **$a^n b^n c^n$, where $n > 0$** ✓
- c) $a^n c^i b^n$, where $n, i > 0$
- d) $c^i a^n b^n$, where $n, i > 0$

CORRECT ANSWER

Answer: B

$a^n b^n c^n$, where $n > 0$

3. The symbols belong to the stack of a PDA

- a) Terminals only
- b) Non-terminals only
- c) States
- d) **Both terminals and non-terminals** ✓

CORRECT ANSWER

Answer: D

Both terminals and non-terminals

4. L is accepted by a PDA where no symbol is necessarily removed from the stack. L is in particular

- a) Context-sensitive
- b) Unrestricted
- c) Context-free
- d) **Regular** ✓

CORRECT ANSWER

Answer: D

Regular

5. Which of the following languages is accepted by a non-deterministic pushdown automaton (PDA) but not by a deterministic PDA?

- a) Always regular
- b) **Never regular** ✓
- c) Always a deterministic context-free language
- d) Always a context-free language

CORRECT ANSWER

Answer: B

Never regular

6. Let L be a context-free language and M a regular language. Then, the language LnM is

- a) $\{a^n b^n c^n \mid n \geq 0\}$
- b) $\{a^l b^m c^n \mid l \neq m \text{ or } m = n\}$
- c) $\{a^n b^n \mid n \geq 0\}$
- d) **$\{a^m b^n \mid m, n \geq 0\}$** ✓

CORRECT ANSWER

Answer: D

$\{a^m b^n \mid m, n \geq 0\}$

7. Which of the following language over $\{a, b, c\}$ is accepted by a deterministic pushdown automata?

- a) **$\{WCW^R \mid w \in (a, b)^*\}$** ✓
- b) $\{WW^R \mid w \in (a, b)^*\}$
- c) $\{a^n b^n c^n \mid n \geq 0\}$
- d) $\{w \mid w \text{ is a palindrome over } \{a, b, c\}\}$

CORRECT ANSWER

Answer: A

$\{WCW^R \mid w \in (a, b)^*\}$

8. The transitional functions of a pushdown automata are given. What is the language accepted by the PDA by the empty store?

- a) $1^n 0 1^n$
- b) $1^n 1^n$
- c) **$1^n 0 1^n \cup \epsilon$** ✓
- d) ϵ

CORRECT ANSWER

Answer: C

$1^n 0 1^n \cup \epsilon$

9. Let L_D be the set of all languages accepted by a PDA by the final state and L_E the set of all languages accepted by the empty stack. Which of the following is true?

- a) **$L_D = L_E$** ✓
- b) $L_D \supset L_E$
- c) $L_E = L_D$
- d) None of the above

CORRECT ANSWER

Answer: A

$L_D = L_E$

10. Let $G = (\{S\}, \{a, b\}, R, S)$ be a context-free grammar where the rule set R is $S \rightarrow a S b \mid S S \mid \epsilon$. Which of the following statements is true?

- a) G is not ambiguous
- b) There exists $x, y \in L(G)$ such that $xy \notin L(G)$
- c) **There is a deterministic pushdown automaton that accepts $L(G)$** ✓
- d) We can find a deterministic finite state automaton that accepts $L(G)$

CORRECT ANSWER

Answer: C

There is a deterministic pushdown automaton that accepts $L(G)$

11. Let N_f and N_p denote the classes of languages accepted by the non-deterministic finite automata and the non-deterministic push-down automata, respectively. Let D_f and D_p denote the classes of languages accepted by the deterministic finite automata and the deterministic push-down automata, respectively. Which one of the following is true?

- a) $D_f \subset N_f$ and $D_p \subset N_p$
- b) $D_f \subset N_f$ and $D_p = N_p$
- c) $D_f = N_f$ and $D_p = N_p$
- d) **$D_f = N_f$ and $D_p \subset N_p$** ✓

CORRECT ANSWER

Answer: D

$D_f = N_f$ and $D_p \subset N_p$

12. Consider the languages: $L_1 = \{ww^R \mid w \in \{0, 1\}^*\}$, $L_2 = \{w\#w^R \mid w \in \{0, 1\}^*\}$, where $\#$ is a special symbol, $L_3 = \{ww \mid w \in \{0, 1\}^*\}$. Which one of the following is true?

- a) L_1 is a deterministic CFL
- b) **L_2 is a deterministic CFL** ✓
- c) L_3 is a CFL, but not a deterministic CFL
- d) L_3 is a deterministic CFL

CORRECT ANSWER

Answer: B

L_2 is a deterministic CFL

13. Which one of the following is false?

- a) There is a unique minimal DFA for every regular language
- b) Every NFA can be converted to an equivalent PDA
- c) The complement of every context-free language is recursive
- d) **Every nondeterministic PDA can be converted to an equivalent deterministic PDA** ✓

CORRECT ANSWER

Answer: D

Every nondeterministic PDA can be converted to an equivalent deterministic PDA

14. Let $M = (K, \Sigma, \Gamma, \Delta, s, F)$ be a pushdown automaton with $K = \{s, f\}$, $F = \{f\}$, $\Sigma = \{a, b\}$, $\Gamma = \{a\}$. Which one of the following strings is not a member of $L(M)$?

- a) aaa
- b) aabab
- c) baaba
- d) **bab** ✓

CORRECT ANSWER

Answer: D

bab

UNIT - III: Turing Machine — Multiple Choice Questions

1. The Turing machine is the machine format of ____ language.

- a) **Type 0** ✓
- b) Type 1
- c) Type 2
- d) Type 3

CORRECT ANSWER

Answer: A

Type 0

2. Which is not a part of the mechanical diagram of the Turing machine?

- a) Input tape
- b) Read-write head
- c) Finite control
- d) **Stack** ✓

CORRECT ANSWER

Answer: D

Stack

3. The difference between the Turing machine and the two-way FA is in the

- a) Input tape
- b) **Read-write head** ✓
- c) Finite control
- d) All of these

CORRECT ANSWER

Answer: B

Read-write head

4. The difference between the Turing machine and the pushdown automata is in the

- a) Head movement
- b) Finite control
- c) Stack
- d) **All of these** ✓

CORRECT ANSWER

Answer: D

All of these

5. Which is not true for the mechanical diagram of the Turing machine
- a) The head moves in both directions
 - b) The head reads as well as writes
 - c) The input string is surrounded by an infinite number of blank in both sides
 - d) **Some symbols are pushed into the stack.** ✓

CORRECT ANSWER

Answer: D

Some symbols are pushed into the stack.

6. In the Turing machine, the transitional function δ is in the form (Q finite states, Σ input alphabets, Γ tape symbol, L=left, R=right, H=halt)
- a) **$Q \times \Gamma \rightarrow (Q \times \Gamma \times \{L, R, H\})$** ✓
 - b) $Q \times \Sigma \rightarrow (Q \times \{L, R, H\})$
 - c) $Q \times \Sigma \rightarrow (Q \times \Sigma \times \{L, R, H\})$
 - d) $Q \times \Gamma \rightarrow (Q \times \Sigma \times \{L, R, H\})$

CORRECT ANSWER

Answer: A

$Q \times \Gamma \rightarrow (Q \times \Gamma \times \{L, R, H\})$

7. Which of the strings is accepted by the Turing machine?
- a) $L = a^n c^m b^n$, where $m, n > 0$
 - b) $L = a^n b^n c^i$, where $n, i > 0$
 - c) $L = a^n b^n c^n$, where $n > 0$
 - d) **All of the above** ✓

CORRECT ANSWER

Answer: D

All of the above

8. Which is not true for the instantaneous description (ID) of the Turing machine?
- a) It remembers the state of the machine
 - b) It remembers the cell currently being scanned by the read-write head
 - c) The contents of all the cells of the tape
 - d) **The content of the cell on which the head was in previously** ✓

CORRECT ANSWER

Answer: D

The content of the cell on which the head was in previously

GATE Questions — Turing Machine

1. Which of the following is not accepted by the PDA but accepted by the two-stack PDA?
- a) $a^n b^n$
 - b) $a^n b^m c^m d^n$
 - c) $a^n b^n c^i$
 - d) **$a^n b^n c^n d^n$** ✓

CORRECT ANSWER

Answer: D

$a^n b^n c^n d^n$

2. Which is true of the following? A finite automata can be changed to an equivalent Turing machine by
- a) Changing its reading head to read-write head
 - b) Adding extra initial and final states with input B, B, R.
 - c) **Changing its i/p to i/p, i/p, R and adding extra initial and final states with input B, B, R.** ✓
 - d) None of the above.

CORRECT ANSWER

Answer: C

Changing its i/p to i/p, i/p, R and adding extra initial and final states with input B, B, R.

3. Consider the following Turing machine $\delta(A, a) \rightarrow (B, a, R)$, $\delta(B, b) \rightarrow (C, b, R)$, $\delta(C, a) \rightarrow (C, a, R)$, $\delta(C, b) \rightarrow (D, b, R)$. D is the final state. The language accepted by the TM is
- a) aba^*
 - b) aba^*ab
 - c) **aba^*b** ✓
 - d) a^*ba

CORRECT ANSWER

Answer: C

aba^*b

4. Consider the following Turing machine $\delta(A, a) \rightarrow (B, a, R)$, $\delta(B, b) \rightarrow (B, b, R)$, $\delta(B, a) \rightarrow (C, a, R)$, $\delta(C, b) \rightarrow (C, b, R)$, $\delta(C, a) \rightarrow (D, a, R)$. D is the final state. The language accepted by the TM is
- a) ab^*ab^*
 - b) **ab^*ab^*a** ✓
 - c) $ab + ab + a$
 - d) $a^*(ba)^*a$

CORRECT ANSWER

Answer: B

ab^*ab^*a

5. Consider the following Turing machine $\delta(A, a) \rightarrow (B, a, R)$, $\delta(A, b) \rightarrow (B, b, R)$, $\delta(B, a) \rightarrow (A, a, L)$, $\delta(B, b) \rightarrow (A, b, L)$, $\delta(B, \text{Blank}) \rightarrow (C, \text{Blank}, R)$. C is the final state. On input 'ab', the machine will
- a) Halt on accepting state
 - b) **Go into infinite loop** ✓
 - c) Crash
 - d) Reach to final state but will not halt.

CORRECT ANSWER

Answer: B

Go into infinite loop

6. When does a Turing machine crash?
- a) If the machine traverses all the inputs without traversing some states
 - b) If it traverses all its states till the input remains
 - c) **If the transitional function is not defined for the present state and the input combination ✓**
 - d) None of these.

CORRECT ANSWER

Answer: C

If the transitional function is not defined for the present state and the input combination

7. A single-tape Turing machine M has two states q_0 and q_1 , of which q_0 is the starting state. The tape alphabet of M is $\{0, 1, B\}$ and its input alphabet is $\{0, 1\}$. Which of the following statements is true about M?
- a) **M does not halt on any string in $(0 + 1)^+$ ✓**
 - b) M does not halt on any string in $(00 + 1)^*$
 - c) M halts on all strings ending in a 0
 - d) M halts on all strings ending in a 1

CORRECT ANSWER

Answer: A

M does not halt on any string in $(0 + 1)^+$

8. Consider the languages L_1 , L_2 , and L_3 : $L_1 = \{0^p 1^q \mid p, q \in \mathbb{N}\}$, $L_2 = \{0^p 1^q \mid p, q \in \mathbb{N} \text{ and } p = q\}$, $L_3 = \{0^p 1^q 1^r \mid p, q, r \in \mathbb{N} \text{ and } p = q = r\}$. Which of the following statements is not true?
- a) Pushdown automata (PDA) can be used to recognize L_1 and L_2
 - b) L_1 is a regular language
 - c) **All the three languages are context free ✓**
 - d) Turing machines can be used to recognize all the languages

CORRECT ANSWER

Answer: C

All the three languages are context free

Other Models of Turing Machines — Multiple Choice Questions

1. A multi-tape Turing machine has
- a) Multiple tape
 - b) Multiple head
 - c) Multiple finite control
 - d) **Multiple tape and head ✓**

CORRECT ANSWER

Answer: D

Multiple tape and head

2. A multi-head Turing machine has
- a) Multiple tape
 - b) **Multiple head ✓**
 - c) Multiple finite control
 - d) Multiple tape and head

CORRECT ANSWER

Answer: B

Multiple head

3. In respect to the k-dimensional Turing machine, a simple Turing machine is
- a) 0 dimensional
 - b) **1 dimensional ✓**
 - c) 2 dimensional
 - d) 3 dimensional

CORRECT ANSWER

Answer: B

1 dimensional

4. A simple Turing machine is
- a) **1-way infinite tape ✓**
 - b) 2-way infinite tape
 - c) No way infinite tape
 - d) none of these

CORRECT ANSWER

Answer: A

1-way infinite tape

5. A Post machine has a
- a) Stack
 - b) Linear list
 - c) **Queue ✓**
 - d) Circular queue

CORRECT ANSWER

Answer: C

Queue

GATE Questions — Other Models of Turing Machines

1. Which of the following is true (A: Multi-tape TM, B: Multi-head TM, C: Non-deterministic TM, D: K-dimensional TM, E: Single tape TM).
- a) $A, B > E$
 - b) $C > E$
 - c) $D > A, B > C > E$
 - d) **$A = B = C = D = E$ ✓**
2. Which of the following has a read only tape?
- a) Single-tape TM
 - b) Multi-tape TM
 - c) **Linear bounded automata ✓**
 - d) None of these.

CORRECT ANSWER

Answer: D

$A = B = C = D = E$

CORRECT ANSWER

Answer: C

Linear bounded automata

3. Which one of the following conversions is impossible?

- a) NTM $\rightarrow$ DTM
- b) Multi-tape TM $\rightarrow$ Single-tape TM
- c) Multi-head TM $\rightarrow$ Single-head TM
- d) **NPDA $\rightarrow$ DPDA** ✓

CORRECT ANSWER

Answer: D

NPDA $\rightarrow$ DPDA

4. The movement of the head is limited to a certain region for which of the following machine

- a) K- dimensional TM
- b) **Linear bounded automata** ✓
- c) Pushdown automata
- d) Two-way infinite tape

CORRECT ANSWER

Answer: B

Linear bounded automata

5. In which of the cases stated is the following statement true? 'For every non-deterministic machine M_1 , there exists an equivalent deterministic machine M_2 recognizing the same language.'

- a) **M_1 is a non-deterministic finite automata** ✓
- b) M_1 is a non-deterministic PDA
- c) M_1 is a non-deterministic TM
- d) For no machine M_1 is the above statement true.

CORRECT ANSWER

Answer: A

M_1 is a non-deterministic finite automata

Undecidable Problems — Multiple Choice Questions

1. The Turing machine accepts

- a) Regular language
- b) Context-free language
- c) Context-sensitive language
- d) **All of these** ✓

CORRECT ANSWER

Answer: D

All of these

2. A language is recursively enumerable if a Turing machine

- a) Halts and accepts
- b) Halts and rejects
- c) loops forever
- d) **performs (a), (b), or (c)** ✓

CORRECT ANSWER

Answer: D

performs (a), (b), or (c)

3. A language L is called decidable (or just decidable), if there exists a Turing machine M which

- a) **Accepts L** ✓
- b) Rejects L
- c) Loops forever on L
- d) performs (a), (b), or (c)

CORRECT ANSWER

Answer: A

Accepts L

4. Find the true statement

- a) A recursively enumerable language is a subset of a recursive language
- b) **A recursive language is a subset of a recursively enumerable language** ✓
- c) Both are equivalent
- d) Both may loop forever on the input to a Turing machine

CORRECT ANSWER

Answer: B

A recursive language is a subset of a recursively enumerable language

5. Which is false for recursive language?

- a) Union of two recursive languages is recursive
- b) Complement of a recursive language is recursive
- c) **Recursive language may loop forever on Turing machine** ✓
- d) String belongs to a recursive language either accepts or rejects on Turing machine.

CORRECT ANSWER

Answer: C

Recursive language may loop forever on Turing machine

6. Find the decidable problem regarding the DFA.

- a) The problem that a set of null strings is accepted by a DFA M
- b) The problem that a string w is accepted by a DFA M
- c) The problem that two DFA, M_1 and M_2 , satisfy the same language
- d) **All of these** ✓

CORRECT ANSWER

Answer: D

All of these

7. Which is true for reducibility?

- a) Converting one problem to another problem.
- b) Converting one solved problem to another unsolved problem.
- c) **Converting one unsolved problem into another solved problem to solve the first problem.** ✓
- d) Converting one unsolved problem to another unsolved problem.

CORRECT ANSWER

Answer: C

Converting one unsolved problem into another solved problem to solve the first problem.

1. Which of the following statements is false?
- a) The halting problem of a Turing machine is undecidable.
 - b) Determining whether a context-free grammar is ambiguous is undecidable.
 - c) Given two arbitrary context-free grammars G_1 and G_2 , it is undecidable whether $L(G_1) = L(G_2)$.
 - d) **Given two regular grammars G_1 and G_2 , it is undecidable whether $L(G_1) = L(G_2)$.** ✓

CORRECT ANSWER

Answer: D

Given two regular grammars G_1 and G_2 , it is undecidable whether $L(G_1) = L(G_2)$.

2. Which of the following is not decidable?
- a) Given a Turing machine M , a string s , and an integer k , M accepts s within k steps
 - b) **The equivalence of two given Turing machines** ✓
 - c) The language accepted by a given finite state machine is not empty.
 - d) The language generated by a context-free grammar is non-empty.

CORRECT ANSWER

Answer: B

The equivalence of two given Turing machines

3. Consider the following decision problems: P_1 : Does a given finite state machine accept a given string; P_2 : Does a given context-free grammar generate an infinite number of strings. Which of the following statements is true?
- a) **Both P_1 and P_2 are decidable** ✓
 - b) Neither P_1 nor P_2 are decidable
 - c) Only P_1 is decidable
 - d) Only P_2 is decidable

CORRECT ANSWER

Answer: A

Both P_1 and P_2 are decidable

4. Consider the following problem X . Given a Turing machine M over the input alphabet Σ , any state q of M and a word $w \in \Sigma^*$, does the computation of M on w visit the state q ?
- a) X is decidable
 - b) **X is undecidable but partially decidable** ✓
 - c) X is undecidable and not even partially decidable
 - d) X is not a decision problem

CORRECT ANSWER

Answer: B

X is undecidable but partially decidable

5. Which of the following is true?
- a) **The complement of a recursive language is recursive** ✓
 - b) The complement of a recursively enumerable language is recursively enumerable
 - c) The complement of a recursive language is either recursive or recursively enumerable
 - d) The complement of a context-free language is context free

CORRECT ANSWER

Answer: A

The complement of a recursive language is recursive

6. L_1 is a recursively enumerable language over Σ . An algorithm A effectively enumerates its words as $w_1, w_2, w_3, \dots$. Define another language L_2 over $\Sigma \cup \{\#\}$ as $\{w_i \# w_j : w_j \in L_1, i < j\}$. Consider: S_1 : L_1 is recursive implies that L_2 is recursive; S_2 : L_2 is recursive implies that L_1 is recursive. Which of the following statements is true?
- a) Both S_1 and S_2 are true
 - b) **S_1 is true but S_2 is not necessarily true** ✓
 - c) S_2 is true but S_1 is not necessarily true
 - d) Neither is necessarily true

CORRECT ANSWER

Answer: B

S_1 is true but S_2 is not necessarily true

7. Consider three decision problems P_1, P_2 , and P_3 . It is known that P_1 is decidable and P_2 is undecidable. Which one of the following is true?
- a) P_3 is decidable if P_1 is reducible to P_3
 - b) P_3 is undecidable if P_3 is reducible to P_2
 - c) **P_3 is undecidable if P_2 is reducible to P_3** ✓
 - d) P_3 is decidable if P_3 is reducible to P_2 's complement

CORRECT ANSWER

Answer: C

P_3 is undecidable if P_2 is reducible to P_3

8. Let L_1 be a recursive language, and let L_2 be a recursively enumerable but not a recursive language. Which one of the following is true?
- a) Complement of L_1 is recursive and complement of L_2 is recursively enumerable.
 - b) **Complement of L_1 is recursive and complement of L_2 is not recursively enumerable.** ✓
 - c) Complement of L_1 and complement of L_2 are recursively enumerable.
 - d) Complement of L_1 is recursively enumerable and complement of L_2 is recursive

CORRECT ANSWER

Answer: B

Complement of L_1 is recursive and complement of L_2 is not recursively enumerable.

9. For $S \in (0 + 1)^*$, let $d(s)$ denote the decimal value of s [e.g., $d(101) = 5$]. Let $L = \{s \in (0 + 1)^* \mid d(s) \bmod 5 = 2 \text{ and } d(s) \bmod 7 \neq 4\}$. Which one of the following statements is true?
- a) L is recursively enumerable, but not recursive
 - b) **L is recursive, but not context free** ✓
 - c) L is context free, but not regular
 - d) L is regular

CORRECT ANSWER

Answer: B

L is recursive, but not context free

10.

Let L_1 be a regular language, L_2 be a deterministic context-free language, and L_3 a recursively enumerable, but not recursive language. Which one of the following statements is false?

a) $L_1 \cap L_2$ is a deterministic CFL

b) $L_3 \cap L_1$ is recursive ✓

c) $L_1 \cup L_2$ is context free

d) $L_1 \cap L_2 \cap L_3$ is recursively enumerable

CORRECT ANSWER

Answer: B

$L_3 \cap L_1$ is recursive

11.

Which of the following problems is undecidable?

a) Membership problem for CFGs.

b) Ambiguity problem for CFGs. ✓

c) Finiteness problem for FSAs

d) Equivalence problem for FSAs.

CORRECT ANSWER

Answer: B

Ambiguity problem for CFGs.

12.

The language $L = \{0^i21^i \mid i \geq 0\}$ over alphabet $\{0, 1, 2\}$ is

a) not recursive

b) recursive and is a deterministic CFL. ✓

c) a regular language.

d) not a deterministic CFL but a CFL.

CORRECT ANSWER

Answer: B

recursive and is a deterministic CFL.

13.

Which of the following are decidable? (i) Whether the intersection of two regular languages is infinite (ii) Whether a given context-free language is regular (iii) Whether two pushdown automata accept the same language (iv) Whether a given grammar is context free

a) (i) and (ii)

b) (i) and (iv) ✓

c) (ii) and (iii)

d) (ii) and (iv)

CORRECT ANSWER

Answer: B

(i) and (iv)

14.

If L and $\text{complement}(L)$ are recursively enumerable, then L is

a) Regular

b) Context free

c) Context sensitive

d) Recursive ✓

CORRECT ANSWER

Answer: D

Recursive

15.

Which of the following statements is false?

a) Every NFA can be converted to an equivalent DFA

b) Every non-deterministic Turing machine can be converted to an equivalent deterministic Turing machine.

c) Every regular language is also a context-free language

d) Every subset of a recursively enumerable set is recursive. ✓

CORRECT ANSWER

Answer: D

Every subset of a recursively enumerable set is recursive.

16.

Let L_1 be a recursive language. Let L_2 and L_3 be the languages that are recursively enumerable but not recursive. Which of the following statements is not necessarily true?

a) $L_2 - L_1$ is recursively enumerable

b) $L_1 - L_3$ is recursively enumerable ✓

c) $L_2 \cap L_1$ is recursively enumerable

d) $L_2 \cup L_1$ is recursively enumerable

CORRECT ANSWER

Answer: B

$L_1 - L_3$ is recursively enumerable

17.

Let $L = L_1 \cap L_2$, where $L_1 = \{a^m b^m c a^n b^n, m, n \geq 0\}$, $L_2 = \{a^i b^j c^k, i, j, k \geq 0\}$. Then, L is

a) **not recursive ✓**

b) regular

c) context-free but not regular

d) recursively enumerable but not context free

CORRECT ANSWER

Answer: A

not recursive

18.

Which of the following problems are decidable? (i) Does a given program ever produce an output? (ii) If L is a context-free language, then, is $\text{complement}(L)$ also context free? (iii) If L is a regular language, then is $\text{complement}(L)$ also regular? (iv) If L is a recursive language, then is $\text{complement}(L)$ also recursive?

a) (i), (ii), (iii), (iv)

b) (i), (ii)

c) (ii), (iii), (iv)

d) (iii), (iv) ✓

CORRECT ANSWER

Answer: D

(iii), (iv)